

GOGEBIC-IRON COUNTY AP, MI, USA

WMO: 727445

Lat: 46.533N

Lon: 90.133W

Elev: 1230

StdP: 14.05

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94926

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -13.8	(c) -8.4	(d) -23.8	(e) 1.5	(f) -11.0	(g) -17.6	(h) 2.2	(i) -6.0	(j) 24.4	(k) 12.9	(l) 22.1	(m) 12.2	(n) 7.9	(o) 250	(p) 0.614

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.2	(c) 85.9	(d) 70.2	(e) 82.2	(f) 68.1	(g) 80.6	(h) 66.8	(i) 72.9	(j) 82.6	(k) 70.8	(l) 80.0	(m) 68.7	(n) 77.2	(o) 10.6	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.9	115.3	79.0	67.3	105.2	76.5	65.5	98.6	75.2	37.3	82.7	35.2	79.4	33.7	78.1	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 22.0	(b) 19.2	(c) 17.4	(e) -21.2	(f) 90.9	(g) 5.2	(h) 3.3	(i) -24.9	(j) 93.3	(k) -28.0	(l) 95.2	(m) -30.8	(n) 97.0	(o) -34.6	(p) 99.4		
(4) 22.0	(5) 19.2	(6) 17.4	(7) -21.3	(8) 76.1	(9) 4.3	(10) 2.0	(11) -24.4	(12) 77.6	(13) -26.9	(14) 78.7	(15) -29.4	(16) 79.8	(17) -32.5	(18) 81.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.4	14.2	16.7	27.1	39.3	52.1	61.5	66.2	63.8	57.1	45.1	32.0	20.0
	DBStd	20.83	12.26	12.61	13.04	10.19	9.24	7.99	6.71	6.60	8.99	9.43	10.21	11.14
	HDD50	4915	1109	932	717	347	84	7	0	0	30	216	544	929
	HDD65	8886	1574	1352	1176	772	413	158	67	102	268	621	989	1394
	CDD50	1776	0	0	6	26	150	351	502	429	244	63	5	0
	CDD65	272	0	0	0	1	15	52	104	66	31	3	0	0
	CDH74	2571	0	0	2	15	196	511	1016	583	228	20	0	0
	CDH80	614	0	0	0	2	53	119	264	125	50	1	0	0
Wind	WSAvg	7.8	8.7	8.4	7.7	7.9	7.7	6.8	6.6	6.4	7.3	8.4	8.8	8.6
Precipitation	PrecAvg	34.9	1.9	1.3	2.0	2.5	3.3	3.9	3.9	3.8	3.7	3.4	2.9	2.1
	PrecMax	46.2	4.3	3.6	4.5	7.9	6.9	7.3	9.2	7.1	7.4	6.7	8.1	5.1
	PrecMin	26.0	0.6	0.2	0.7	0.5	0.6	0.9	1.0	0.7	1.2	1.3	0.6	0.3
	PrecStd	4.8	1.0	0.7	1.0	1.3	1.5	1.6	1.9	1.7	1.6	1.5	1.5	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.3	50.4	66.0	75.3	86.0	88.2	90.9	88.3	86.3	77.2	63.3	46.3
		MCWB	36.4	42.9	56.3	56.8	65.4	72.0	74.1	72.1	70.4	61.6	51.4	40.2
	2%	DB	37.2	44.9	56.6	68.2	80.1	83.4	86.0	82.7	80.5	71.5	55.5	41.7
		MCWB	33.4	38.3	46.9	51.6	61.8	68.9	71.2	69.1	66.7	58.7	47.9	36.4
	5%	DB	34.2	38.8	51.7	63.1	73.4	80.5	82.1	80.7	75.5	65.6	51.9	37.1
		MCWB	31.8	33.8	43.5	48.4	58.8	65.9	69.0	67.9	64.9	57.1	44.5	33.8
	10%	DB	30.3	34.4	45.5	56.9	69.9	75.8	80.5	77.0	72.4	61.1	46.2	34.6
		MCWB	28.3	31.0	38.7	45.3	56.1	63.1	67.6	65.9	62.6	52.6	40.9	32.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.5	43.3	58.1	59.4	70.8	73.8	76.2	74.6	71.5	66.6	54.4	43.0
		MCDB	41.9	50.3	64.6	72.0	81.5	85.2	86.2	83.8	81.8	72.4	60.0	45.6
	2%	WB	34.4	38.6	48.8	53.5	64.6	70.6	73.3	71.8	69.0	60.6	49.6	37.1
		MCDB	37.4	44.0	55.6	65.7	74.0	81.1	83.0	80.1	77.4	68.0	55.9	40.2
	5%	WB	31.7	34.2	43.6	49.6	60.9	67.9	71.1	69.4	66.7	57.0	45.1	34.4
		MCDB	34.2	38.6	50.7	60.5	70.6	76.7	80.4	77.7	74.2	65.1	49.7	37.2
	10%	WB	28.4	31.0	38.7	46.3	57.6	65.3	68.8	67.5	64.1	53.2	41.4	32.1
		MCDB	30.1	34.2	44.8	56.6	68.0	73.6	77.2	75.9	70.7	60.1	46.6	34.7
Mean Daily Temperature Range	5% DB	MDBR	14.3	17.7	19.1	19.6	22.1	21.5	21.2	20.7	19.5	16.1	13.1	12.6
		MCDBR	15.8	20.0	22.3	29.7	28.9	24.3	23.7	23.4	22.3	22.0	20.4	15.5
		MCWBR	13.1	15.0	14.8	17.4	15.3	13.3	12.9	13.1	13.2	13.8	14.6	12.3
	5% WB	MCDBR	15.0	18.1	20.7	26.8	24.2	21.2	21.3	20.8	19.7	19.6	18.7	13.7
		MCWBR	12.6	13.8	14.6	17.0	14.3	13.0	12.5	12.9	12.6	14.1	14.3	11.3
Clear-Sky Solar Irradiance	taub	0.244	0.266	0.298	0.331	0.361	0.362	0.363	0.360	0.344	0.308	0.285	0.255	
	taud	2.435	2.366	2.392	2.431	2.402	2.423	2.450	2.471	2.500	2.576	2.539	2.461	
	Ebn at Noon	279	292	296	294	287	286	284	281	277	274	260	261	
	Edh at Noon	21	28	32	34	36	36	34	32	29	23	19	18	
All-Sky Solar Radiation	RadAvg	409	733	1129	1430	1695	1836	1918	1603	1210	732	417	292	
	RadStd	40	75	99	149	129	129	79	102	69	88	53	33	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (35 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page